Sumon Basak

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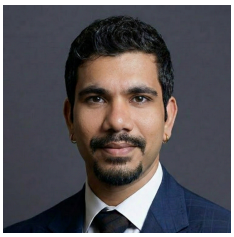

Education

Bar-Ilan University **Dec 2024 - present**
Doctor of Philosophy *Ramat Gan, Israel*

Relevant courses: Data Processing, Analysis and Visualization in Python for Chemistry (84642-01), Computational Chemistry (84328-01), Advanced Simulation Methods for Chemistry, Biology, and Material Science (84992-01)

Banaras Hindu University **Jul 2018 - Dec 2020**
Master of Science *Varanasi, India*

Relevant courses: Computer Applications in Chemistry (CHM-401, 410).

University of Calcutta **Jul 2015 - Jun 2018**
Bachelor of Science. Ranked 2nd. *Kolkata, India*

Chemistry (honors) with Physics (minor) & Mathematics (minor). Received *Nanda Kishore Mukherjee Memorial Prize* for 2nd rank at the department.

Projects

1. **Terpene Synthase Docking**: Dock reactants, reaction intermediates, and products with EnzyDock at the active site of terpene synthases to analyse feasible poses and profile product distribution.
   Relevant Courses: Advanced Python, Computational Chemistry

2. **QD Surface Coating**: Simulate $CsPbBr_3$ quantum dots with a set of ligands and solvents using GROMACS. Model QDs are coated with ligands to compute ligand binding affinity and surface coverage further, and DFT is used to tune the band gap with CP2K.

3. **Enzyme classification**: Develop an ML model to classify protein sequences into 7 parent classes of enzymes. Prepare a data set of sequences from UniProt, filter out homologous entries with sequence identity >50% with CD-HIT, tokenize extracted n-gram features with natural language processing methods, and validate the model through the jackknife test.
   Relevant Courses: Machine Learning A−Z

4. **Crystal growth**: Simulate water-ice phase transition on different planes of hexagonal ice (temp. range 254−272K) deploying GROMACS. From simulation trajectories, calculate the Steinhardt order parameter (q6), track transrotational movement of water on the ice surface, and analyze the influence of neighbors on water crystallization.
   Relevant Courses: Classical Molecular Dynamics, Complete Python Bootcamp (Udemy)

Research Experience

Bar-Ilan University **Dec 2024 - present**
MSCA-ITN doctoral student. *Ramant Gan, Israel*

Advisor: Prof. Dan-Thomas Major
- Molecular docking of terpene synthase products and rational design of novel terpene synthase variants.

Indian Institute of Technology Hyderabad **Aug 2024 - Oct 2024**
Project associate. *Hyderabad, India*

Advisor: Prof. Arup Mahata
- Atomistic simulation of coating perovskite quantum dots with organic ligands and tuning bandgap with DFT calculations.

Indian Association for the Cultivation of Science **Jul 2023 - Jun 2024**
Project associate. *Kolkata, India*

Advisor: Prof. Biman Jana

- Atomistic simulation of water-ice phase transition and machine learning-led classification of enzymes from k-mer fragments of sequences.

Università degli Studi di Perugia **Sep 2021 - Mar 2023**
Student researcher. *Perugia, Italy*

Advisor: Prof. Luigi Vaccaro
- Preparation of optoelectronically-active organoborazine material through a bottom-up approach.

Banaras Hindu University **Jan 2020 - May 2020**
Master's thesis student. *Varanasi, India*

Advisor: Prof. Satyen Saha
- Raman spectroscopic study of xanthene-type skeletons.

Indian Association for the Cultivation of Science **Dec 2019**
Winter intern. *Kolkata, India*

Advisor: Prof. Ayan Datta
- *Ab initio* investigation of H-isotope separation through C_2N-h2D material using VASP.

Indian Institute of Technology Bombay **May 2019 - Jul 2019**
Summer intern. *Mumbai, India*

Advisor: Prof. Debabrata Maiti
- Multi-step syntheses pharmacophoric azole skeletons incorporating Pd-catalyzed aromatic C−H functionalization.

Technical Expertise

- **Simulation packages**: GROMACS, CHARMM, Gaussian, PyMOL, ORCA, QCHEM, VESTA, Chemcraft, VMD
- **Programming languages**: Python, C++, Bash
- **Programming packages**: MDAnalysis, MDTraj, Scikit-learn, TensorFlow, Numpy, Pandas, Matplotlib, Seaborn, Multiprocessing, Biopython
- **Operating systems**: Windows, Linux (Ubuntu), MacOS
- **Other software**: MS Office, Overleaf, Git

Scholarships

MSCA-ITN Fellowship **2025 - 2027**
Got selected to become a doctoral candidate within the MSCA-ITN framework of ModBioTerp. This fellowship is funding first three year of the 4-year long doctoral degree.

SHE-INSPIRE Scholarship **2015 - 2020**
Stood 1st in 12th grade in the school. Ranked amongst the top 1 percentile students of the country and received a full govt. scholarship for the entire B.Sc.-M.Sc. study.

Swami Vivekananda MCM Scholarship **2013 - 2015**
Ranked 2nd in 10th grade in the school and received a full scholarship to study in 11-12 grade.

Publications

1. Dhibar, S.‡, **Basak, S.**‡, and Jana, B. Accurate Enzyme Classification with SOLVE: Integrating Ensemble Classifier and k-mer Tokenization. **Chemical Science** 16(39), pp. 18438. (‡*both authors contributed equally*)
2. Marchionni, D., **Basak, S.**, Khodadadi, A.N., Marrocchi, A. and Vaccaro, L., 2023. Synthesis and Applications of Organic Borazine Materials. **Advanced Functional Materials**, 33(49), pp. 2303635.
3. **Basak, S.**, Dutta, S. and Maiti, D., 2021. Accessing C2-Functionalized 1, 3-(Benz)azoles through Transition Metal-Catalyzed C-H Activation. **Chemistry-A European Journal**, 27(41), pp. 10533.
4. **Basak, S.**, Biswas, J.P. and Maiti, D., 2021. Transition-Metal-Catalyzed C–H Arylation Using Organoboron Reagents. **Synthesis**, 53(18), pp. 3151.

Languages

- **English** (*C1*)
  Scored 111 out of 120 in TOEFL. Conducted by ETS.
- **Italian** (*A2*)
  Scored 25 out of 30. Conducted by CLA at Università degli Studi di Perugia.
- **Bengali** (*Native proficiency*)
- **Hindi** (*Native proficiency*)

Other Activities

- Chemiquiz (**2018**). Teamed with 2 other classmates and ranked 4th in the competition.
- Rotary Swarnakana Merit Test (**2012 - 2014**). Ranked in top 3 thrice.